Chapter 1 · Energy Causes Change

Teacher Key · Guided Notes

Name: _____ **Homeroom:** ○ 4A ○ 4B ○ 4C ○ 4D

KEY *Bold brown = exact answer · Italic gray = sample / acceptable response.*

LANGUAGE NOTE *The Reader says "evidence, or proof." Teach and use the word evidence (it is in the Big Question). If a student writes proof in the N1 blank, accept it and restate: "Yes — we say evidence."*

BQ Where can we observe evidence of energy causing change?

My first thinking

Sample: "A pencil gets shorter and sharper." (Accept any observed change.)

V1 VOCABULARY — ENERGY & CHANGE

Reader p. 1

energy — the ability to cause **change** .

change — to become **different** .

N1 ENERGY CAUSES CHANGE

Reader pp. 1–2

Energy has the ability to cause change.

When you see a change, you are seeing **evidence** of energy.

ENERGY → **CHANGE** → **EVIDENCE**

N2 ENERGY OCCURS IN DIFFERENT FORMS

Reader p. 3

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	*a ball rolls across the floor*
sound	*a drum makes a noise*
light	*a flashlight lights up the room*
heat	*a warm mug warms my hands*
electrical energy	*a fan spins when plugged in*

V2 VOCABULARY – STORED ENERGY

Reader p. 4

stored energy – energy that has the ability to cause change at a **later** time.

N3 OBJECTS CAN STORE ENERGY

Reader pp. 4–5

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	*the flashlight lights up*
stretched rubber band	*it snaps and makes a sound*
raised object (apple)	*it falls down*

Objects also have stored energy because of their **position** .

N4 RECOGNIZING EVIDENCE

Reader pp. 1–5

Draw a change you observed. Show what it looked like BEFORE and AFTER.

BEFORE	AFTER
Sample: a dull pencil, long, rounded tip	*Sample: shorter pencil, sharp tip, shavings*

What changed?

Sample: "The pencil got shorter and sharper." (Accept any observed change.)

CHECK YOUR THINKING

How do we know energy is involved if we cannot see energy itself?

Start with: "We know because we see …"

Sample: "We know because we see a change, like a ball starting to move or a light turning on." (Accept any response connecting a change to evidence of energy.)

BQ · NOW I THINK Where can we observe evidence of energy causing change?

Sample: "We see evidence of energy when things move, make sound, light up, heat up, or change shape." (Accept any change named as evidence.)